Midpoint	$\vec{OM} = \dfrac{\vec{OA} + \vec{OB}}{2}$	Pg 2
Dot Product	$a \cdot b = \lvert a\rvert \lvert b\rvert \cos\theta$	Pg 2
Angle btwn Vectors	$\cos\theta = \dfrac{a \cdot b}{\lvert a\rvert \lvert b\rvert}$	Pg 3
Perp Vectors	$a \cdot b = 0$	Pg 3
Length of Projection	$\lvert c\rvert = \dfrac{\lvert a \cdot b\rvert}{\lvert b\rvert}$	Pg 3
Vec Eqn of Line	$l : r = a + \lambda b$	Pg 4
Cartesian Eqn of Line	$\dfrac{x - x_1}{a} = \dfrac{y - y_1}{b} = \dfrac{z - z_1}{c}$	Pg 5
Angle btwn Lines	$\cos\theta = \dfrac{\lvert b_2 \cdot b_2\rvert}{\lvert b_1\rvert \lvert b_2\rvert}$	Pg 5
Cross Product	$a \times b = \lvert a\rvert \lvert b\rvert \sin\theta$	Pg 6
Area of Parallelogram	$\lvert a \times b\rvert$	Pg 6
Area of Triangle	$\frac{1}{2}\lvert a \times b\rvert$	Pg 7
Perp Dist From Point to Line	$\lvert c\rvert = \dfrac{\lvert a \times b\rvert}{\lvert b\rvert}$	Pg 8
Scalar Eqn of Plane	$r \cdot n = a \cdot n$	Pg 8
Cartesion Eqn of Plane	$ax + by + cz = d$	Pg 9
Vector Eqn of Plane	$r = a + \lambda b + \mu c$	Pg 9

Shortest Dist from Point to Plane	$\vec{PF} = \dfrac{\lvert \vec{AP} \cdot n\rvert}{\lvert n\rvert}$	Pg 10
Angle Btwn Line and Plane	$\sin\theta = \dfrac{\lvert n \cdot d\rvert}{\lvert n \cdot d\rvert}$	Pg 11
Dist Btwn 2 Parallel Planes	$\dfrac{d_1 - d_2}{\lvert n\rvert}$	Pg 12
Intersecting Planes	$n_1 \times n_2 = l$	Pg 13
Angle Btwn 2 Planes	$\cos\theta = \dfrac{\lvert n_1 \cdot n_2\rvert}{\lvert n_1\rvert \lvert n_2\rvert}$	Pg 14
Perp Planes	$n_1 \cdot n_2 = 0$	Pg 14
Shotest Dist btwn 2 Skew lines	$\dfrac{\lvert \vec{A_1A_2} \cdot (b_1 \times b_2)\rvert}{\lvert b_1 \times b_2\rvert}$	Pg 15

$$D_u f = \hat{\underset{\sim}{u}} \cdot \nabla f \big|_{\underset{\sim}{x} = x_0}$$

Tangent plane

general pos.vec

$\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$

$$f(x,y) \approx f(x_0, y_0) + [\underset{\sim}{r} - \underset{\sim}{r_0}] \cdot \nabla f \big|_{r_0 = x_0}$$

$$z = f(x_0, y_0) + [\underset{\sim}{r} - \underset{\sim}{r_0}] \cdot \nabla f \big|_{r_0 = x_0}$$

Critical points $\longrightarrow D_u f = 0$

1. Max $\longrightarrow D_u^2 f < 0$ OR $\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2} < 0$ AND $\det(H) = \left(\frac{\partial^2 f}{\partial x^2}\right)\left(\frac{\partial^2 f}{\partial y^2}\right) - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 > 0$
2. Min $\longrightarrow D_u^2 f > 0$ OR $\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2} > 0$ AND $\det(H) = \left(\frac{\partial^2 f}{\partial x^2}\right)\left(\frac{\partial^2 f}{\partial y^2}\right) - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 > 0$
3. Saddle $\longrightarrow \det(H) = \left(\frac{\partial^2 f}{\partial x^2}\right)\left(\frac{\partial^2 f}{\partial y^2}\right) - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 < 0$

objective constraint

Lagrange, $\nabla f = \lambda \nabla g$

Tangent vec. $\frac{d}{dt} \underset{\sim}{r}(t)$

$$\frac{d}{dt} \underset{\sim}{y} = A \underset{\sim}{y}$$

$$y = c_1 e^{\lambda_1 x} \underset{\sim}{v_1} + c_2 e^{\lambda_2 x} \underset{\sim}{v_2} + \ldots$$

e.val e.vec

$$\sinh^{-1} x = \ln\left(x + \sqrt{x^2 + 1}\right)$$
$$\cosh^{-1} x = \ln\left(x + \sqrt{x^2 - 1}\right)$$
$$\tanh^{-1} x = \frac{1}{2} \ln\left[\frac{1 + x}{1 - x}\right]$$

$$\lim_{n \to \infty} \left|\frac{a_{n+1}}{a_n}\right| = L$$

1. $L < 1$, converge
2. $L > 1$, diverge
3. $L = 1$, inconclusive

Maclauren : $f(x) = f(0) + [f'(0)]x + \left[\frac{f''(0)}{2!}\right]x^2 + \left[\frac{f'''(0)}{3!}\right]x^3 + \ldots$

Taylors : $f(x) = f(x_0) + [f'(x_0)](x - x_0) + \left[\frac{f''(x_0)}{2!}\right](x - x_0)^2 + \left[\frac{f'''(x_0)}{3!}\right](x - x_0)^3 + \ldots$

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x)}{g'(x)}$$

P3 Differentiation

Differentiation of standard functions

Functions (y)	Derivatives $\left(\frac{dy}{dx}\right)$	
constant, k	0	
x^n	nx^{n-1}	
e^x	e^x	
a^x +ve constant	$a^x \ln a$	$\frac{d}{dx}(3^x) = 3^x \ln 3$
$\ln x$	$\frac{1}{x}$	$\frac{d}{dx}[\ln(x^2+3x+5)] = \frac{1}{x^2+3x+5}(2x+3)$
$\sin x$	$\cos x$	$\frac{d}{dx}(\sin 2x) = (\cos 2x)(2) = 2\cos 2x$
$\cos x$	$-\sin x$	
$\tan x$	$\sec^2 x$	$\frac{d}{dx}(\tan 3x) = (\sec^2 3x)(3) = 3\sec^2 3x$
$\tan^{-1} x$	$\frac{1}{1+x^2}$	$\frac{d}{dx}(\tan^{-1}(3x)) = \frac{1}{1+(3x)^2}(3) = \frac{3}{1+9x^2}$

Differentiation of common functions

Functions (y)	Derivatives $\left(\frac{dy}{dx}\right)$
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$
$\frac{1}{x}$	$-\frac{1}{x^2}$
e^{kx}	ke^{kx}
$\ln(ax+b)$	$\frac{a}{ax+b}$
$\sin kx$	$k\cos kx$
$\cos kx$	$-k\sin kx$
$\tan kx$	$k\sec^2 kx$

Differentiating Hyperbolic Functions

1. The standard results are:

(a) $\frac{d}{dx}(\cosh x) = \sinh x$

(b) $\frac{d}{dx}(\sinh x) = \cosh x$

(c) $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$

$\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$

$\frac{d}{dx}(\operatorname{cosech} x) = -\operatorname{cosech} x \coth x$

$\frac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x$

Recall

$\frac{d}{dx}(\cos x) = -\sin x$

$\frac{d}{dx}(\sin x) = \cos x$

$\frac{d}{dx}(\tan x) = \sec^2 x$

$\frac{d}{dx}(\sec x) = \sec x \tan x$

$\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

(a) $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$

(b) $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{x^2+1}}$

(c) $\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$

There are **three types of solution to this quadratic equation** (two distinct real roots / two equal real roots / two complex roots which are conjugates of each other), and **they each lead to different types of solution for the differential equation**.

Value of discriminant	***Type of roots for the auxiliary equation***	***Complementary function***
$b^2 - 4ac > 0$	Real, distinct roots, λ_1 and λ_2	$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$
$b^2 - 4ac = 0$	Real, repeated root, λ	$y = (A + Bx)e^{\lambda x}$ [Just memorise]
$b^2 - 4ac < 0$	Complex roots: **(a)** $\alpha \pm \beta i$, with $\alpha \neq 0$ **(b)** $\pm \beta i$ (i.e. **purely imaginary** roots)	 **(a)** $y = e^{\alpha x}(P\cos\beta x + Q\sin\beta x)$ **(b)** $y = P\cos\beta x + Q\sin\beta x$

4. The table below is a **guide** to **what trial function to use** for the differential equation

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x)$$

Right-hand side **(i.e. the $f(x)$)**	***Trial function***
Linear function	$Px + Q$
Polynomial of degree n	$P_n x^n + P_{n-1}x^{n-1} + \cdots + P_1 x + P_0$ (typically we will just use letters P, Q, R, etc rather than using subscripts on the same letter)
Trigonometric function involving $\cos kx$ and/or $\sin kx$	$P\cos kx + Q\sin kx$
Exponential function involving e^{kx}	Pe^{kx}
Sum of different functions	Sum of matching functions

You should note that this method finds a simple particular integral, but actually there are infinitely many possible particular integrals, which can be constructed by adding on any term from the complementary function.

Multiplication of exponential func.

(Polynomial of deg n) e^{kx}	$y_p(x) = [P_n x^n + P_{n-1}x^{n-1} + \dots + P_0]e^{kx}$
(Trig involving $\cos mx$ and/or $\sin mx$)(e^{kx})	$y_p(x) = [P\cos(mx) + Q\sin(mx)]e^{kx}$
(Polynomial deg n)(Trigo of $\cos mx$ and/or $\sin mx$)	$y_p(x) = (Ax^n + Bx^{n-1} \dots + C)\cos(mx) + (Dx^n + Ex^{n-1} \dots + F)\sin(mx)$

TABLE 4.4.1 Trial Particular Solutions

$g(x)$	Form of y_p
1. 1 (any constant)	A
2. $5x + 7$	$Ax + B$
3. $3x^2 - 2$	$Ax^2 + Bx + C$
4. $x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + E$
5. $\sin 4x$	$A \cos 4x + B \sin 4x$
6. $\cos 4x$	$A \cos 4x + B \sin 4x$
7. e^{5x}	Ae^{5x}
8. $(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9. x^2e^{5x}	$(Ax^2 + Bx + C)e^{5x}$
10. $e^{3x} \sin 4x$	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
11. $5x^2 \sin 4x$	$(Ax^2 + Bx + C) \cos 4x + (Ex^2 + Fx + G) \sin 4x$
12. $xe^{3x} \cos 4x$	$(Ax + B)e^{3x} \cos 4x + (Cx + E)e^{3x} \sin 4x$